# Homogenization: Triadic Partition, Cube Measure, Cube Average, and Telescoping Increment Lemmas in Lean 4

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**Abstract**

We present a complete, zero-`sorry` Lean 4 (Mathlib) formalization of the multi-scale triadic cube hierarchy on $\mathbb{R}^d$ ('Eucl d := Fin d -> R') required for quantitative stochastic and PDE homogenization arguments. Specifically, inside the namespace `Homogenization`, we construct the half-open triadic cube $\square_m(z) = \prod_{i=1}^{d} [z_i - \frac{1}{2}3^m, z_i + \frac{1}{2}3^m)$, prove its exact Lebesgue volume $\mathrm{vol}(\square_m(z)) = 3^{md}$ via Fubini–Tonelli product integration over `Fin d`, define the normalized cube average $\mathrm{cubeAvg}(m, z, g) = 3^{-md} \int_{\square_m(z)} g\, dx$, establish linearity and constant-function preservation for $L^1_{\mathrm{loc}}$ fields, and close the telescoping multi-scale increment identity

$$\mathrm{cubeAvg}(m+K, z, g) - \mathrm{cubeAvg}(m, z, g) = \sum_{k=0}^{K-1} \big(\mathrm{cubeAvg}(m+k+1, z, g) - \mathrm{cubeAvg}(m+k, z, g)\big)$$

together with the corresponding normed triangle inequality bound.

## 1 Provenance and Verification Record

- **Author:** Benjamin Stanley Frohman (`ORCID: 0009-0006-7068-3718`).
- **Origin Provenance Timestamp:** Monday, July 6, 2026, 00:49:25 CDT (`2026-07-06T05:49:25.627Z`, Google Drive Archive ID `1N6WfgOBJkPciRYWIQHJqAa2cmuWdu0yH`).
- **Formalization Artifact:** `TriadicHomogenization.lean` (183 lines, zero `sorry`, zero custom axioms).

## 2 Mathematical Statements Formalized in Lean 4

**Definition 1** (Half-Open Triadic Cube)**.** *For $d \in \mathbb{N}$, scale exponent $m \in \mathbb{N}$, and center $z \in \mathbb{R}^d$, the triadic cube* $\mathrm{cube}(m, z) \subset \mathbb{R}^d$ *is defined coordinatewise by*

$$\mathrm{cube}(m, z) = \bigcap_{i \in \mathrm{Fin}\, d} \left\{ x \in \mathbb{R}^d \;\middle|\; z_i - \frac{3^m}{2} \le x_i < z_i + \frac{3^m}{2} \right\}.$$

**Lemma 2** (Exact Lebesgue Measure of the Triadic Cube – `measure_cube` & `measureReal_cube_eq`)**.** *For every $d, m \in \mathbb{N}$ and $z \in \mathbb{R}^d$, the set* $\mathrm{cube}(m, z)$ *is measurable and its Lebesgue volume satisfies*

$$\mu(\mathrm{cube}(m, z)) = (3^m)^d > 0, \qquad \mu_{\mathbb{R}}(\mathrm{cube}(m, z)) = 3^{md} > 0.$$

**Lemma 3** (Normalized Cube Average Properties – `cubeAvg_add`, `cubeAvg_smul`, `cubeAvg_const`)**.** *Let* $\mathrm{cubeAvg}(m, z, g) := \mu_{\mathbb{R}}(\mathrm{cube}(m, z))^{-1} \int_{x \in \mathrm{cube}(m,z)} g(x)\, d\mu(x)$*. Then:*

1. *For $g_1, g_2$ integrable on* $\mathrm{cube}(m, z)$*,* $\mathrm{cubeAvg}(m, z, g_1+g_2) = \mathrm{cubeAvg}(m, z, g_1)+\mathrm{cubeAvg}(m, z, g_2)$*.*
2. *For $c \in \mathbb{R}$,* $\mathrm{cubeAvg}(m, z, c \cdot g) = c \cdot \mathrm{cubeAvg}(m, z, g)$*.*
3. *For constant $c \in E$,* $\mathrm{cubeAvg}(m, z, x \mapsto c) = c$*.*

**Theorem 4** (Multi-Scale Telescoping Increment Identity and Norm Bound – `telescoping_increments` & `norm_Increment_le_sum`)**.** *For every $m, K \in \mathbb{N}$, $z \in \mathbb{R}^d$, and field $g : \mathbb{R}^d \to E$:*

$$\mathrm{cubeAvg}(m+K, z, g) - \mathrm{cubeAvg}(m, z, g) = \sum_{k=0}^{K-1} \big(\mathrm{cubeAvg}(m+k+1, z, g) - \mathrm{cubeAvg}(m+k, z, g)\big),$$

*and consequently*

$$\big\|\mathrm{cubeAvg}(m+K, z, g) - \mathrm{cubeAvg}(m, z, g)\big\| \le \sum_{k=0}^{K-1} \big\|\mathrm{cubeAvg}(m+k+1, z, g) - \mathrm{cubeAvg}(m+k, z, g)\big\|.$$

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